Part-of-Speech Tagging and Hidden Markov Model

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Part of Speech Tagging

- Given a PoS tag set $T = [t_1, t_2, \cdots, t_m]$and a sentence $S = [w_1, w_2, \cdots, w_n]$
  - Assign each word w_i a tag t_j that defines its grammatical class.

He	ate	an	apple	.
PRP	VBD	DT	NN	.

Part of Speech Tagging

- A word can have more than one PoS tag.

He_PRP went_VBD to_TO the_DT *park_NN* in_IN a_DT car_NN ._.

He_PRP went_VBD to_TO *park_VB* the_DT car_NN in_IN the_DT shed_NN ._.

- Context can help in disambiguation, but not always!

I *bank*^{1} on the *bank*^{2} by the river *bank*^{3}. $\longrightarrow$ *bank*^{1}_Verb
bank^2_Noun
bank^3_Noun

Part of Speech

- Grouping of words into equivalence classes *w.r.t.* the role they play in the syntactic structure.
  - Contains information about the word and its neighbours, e.g., an article always follows a noun/ noun phrase.

- Traditionally,
  - Noun
  - Verb
  - Adjective - modifies noun
  - Adverb - modifies verb
  - Pronoun - refers to a noun
  - Preposition
  - Conjunction
  - Interjection

Content words (Open class)
- Semantically rich word
- Not-fixed

Functional words (Closed class)
- Important to bind the sentence
- Usually, have less information
- Fixed and Limited

Part of Speech - Open class

- Noun
  - Proper
  - Common (count and mass)
- Verb
  - Main
    - Present, Past, Past Participle, Gerund, 3rd-person-sg (-s form)
  - Auxiliary (Closed class)
  - Copula (Closed class)
- Adjective (Comparative, Superlative, ..)
- Adverb
  - Directional/Locative - here, there, etc.
  - Degree - extremely, very
  - Manner - slowly, delicately
  - Temporal - yesterday, Monday, etc.

Part of Speech - Closed class

- Verb
    - Auxiliary - marks certain semantic (tense, action required) features of a main verb
    - Copula - connects subjects with nominals/adjectives

      He *is* a boy. (copula) vs He *is* playing. (auxiliary)
- Pronouns
    - Personal
    - Possessive
    - Wh-pronouns
- Particle
    - resembles a preposition/adverb but usually combine with a verb to form a phrasal verb, e.g., went on, marked off, etc.
- Preposition, Conjunction, Determiners, Numerals, Interjections, Greetings, Negation,

Part of Speech

Penn Tree-bank tagset
45 tags

Tag	Description	Example	Tag	Description	Example
CC	Coordin. Conjunction	*and, but, or*	SYM	Symbol	+,%, &
CD	Cardinal number	*one, two, three*	TO	"to"	*to*
DT	Determiner	*a, the*	UH	Interjection	*ah, oops*
EX	Existential 'there'	*there*	VB	Verb, base form	*eat*
FW	Foreign word	*mea culpa*	VBD	Verb, past tense	*ate*
IN	Preposition/sub-conj	*of, in, by*	VBG	Verb, gerund	*eating*
JJ	Adjective	*yellow*	VBN	Verb, past participle	*eaten*
JJR	Adj., comparative	*bigger*	VBP	Verb, non-3sg pres	*eat*
JJS	Adj., superlative	*wildest*	VBZ	Verb, 3sg pres	*eats*
LS	List item marker	*1, 2, One*	WDT	Wh-determiner	*which, that*
MD	Modal	*can, should*	WP	Wh-pronoun	*what, who*
NN	Noun, sing. or mass	*llama*	WP$	Possessive wh-	*whose*
NNS	Noun, plural	*llamas*	WRB	Wh-adverb	*how, where*
NNP	Proper noun, singular	*IBM*	$	Dollar sign	*$*
NNPS	Proper noun, plural	*Carolinas*	#	Pound sign	*#*
PDT	Predeterminer	*all, both*	"	Left quote	(' or ")
POS	Possessive ending	*'s*	"	Right quote	(' or ")
PP	Personal pronoun	*I, you, he*	(	Left parenthesis	([, (, {, <)
PP$	Possessive pronoun	*your, one's*	)	Right parenthesis	( ],), }, >)
RB	Adverb	*quickly, never*	,	Comma	,
RBR	Adverb, comparative	*faster*	.	Sentence-final punc	(. ! ?)
RBS	Adverb, superlative	*fastest*	:	Mid-sentence punc	(: ; ... – -)
RP	Particle	*up, off*			

Ambiguity in POS tag

Unambiguous (1 tag)	**35,340**	
Ambiguous (2-7 tags)	**4,100**	
2 tags	3,760	
3 tags	264	
4 tags	61	
5 tags	12	
6 tags	2	
7 tags	1	(“still”)

- Most words in English are unambiguous
  - Almost, 88.5% (Brown corpus)
  - If no ambiguity, PoS tagging is simple
    - learn a *table* of words and its corresponding tags.

- However, most common words are ambiguous.
  - Almost, 40% (tokens in Brown corpus)

- Disambiguation
  - Look for the contextual information, i.e., *look-back* or *look-ahead*.

Rules for disambiguating “present”

- Data for Present.
    - He gifted me the/a/this/that **present_NN**.
    - They **present_VB** innovative ideas.
    - He was **present_JJ** in the class.

- For Present_NN (*look-back*)
    - If present is preceded by determiner (the/a) or demonstrative (this/that),
        - then POS tag will be noun.
    - Does this rule guarantee 100% precision and 100% recall?
        - False positive:
            - *The **present_ADJ** case is not convincing.* (Adjective preceded by “the”)
        - False negative:
            - ***Present** foretells the future.* (Noun but not preceded by “the”)

Rules for disambiguating “present”

- For Present_NN (*look-back* and *look ahead*)
  - If present is preceded by determiner (the/a) or demonstrative (this/that) or followed by a verb,
    - then POS tag will be noun.
    - E.g.
      - **Present_NN** foretells the future.
  - Does this rule guarantee 100% precision and 100% recall?

Illustration of PoS tagging

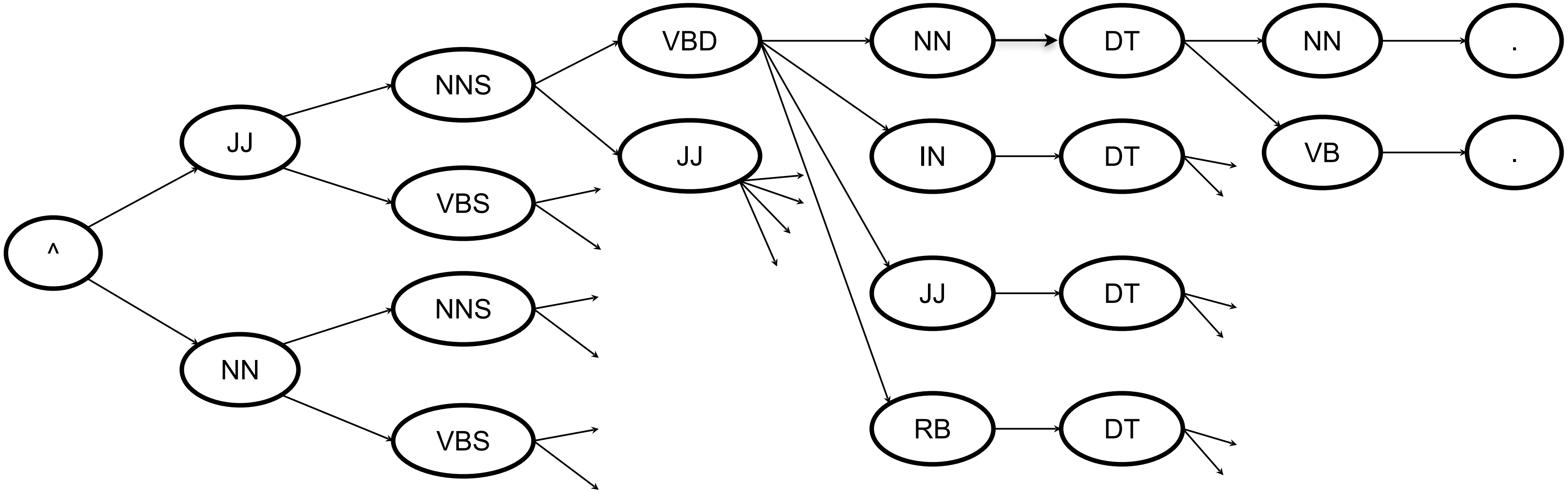
An example

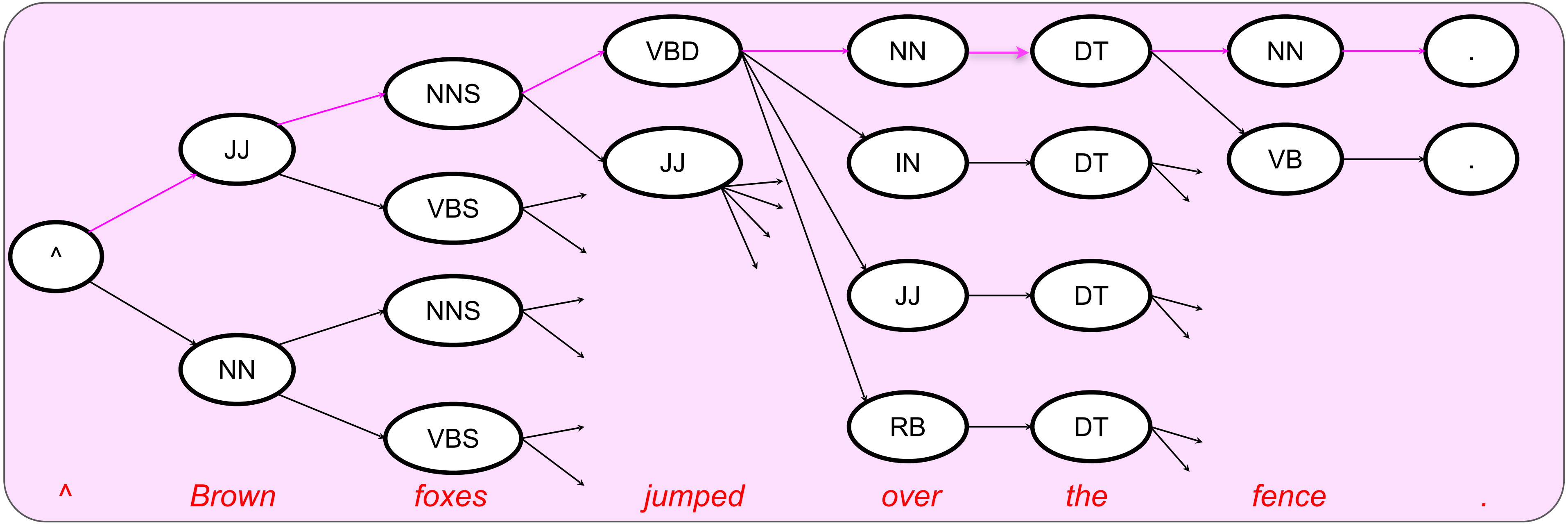
- S = Brown foxes jumped over the fence.
- T = ??

- Tag set = [NN, NNS, VB, VBS, VBD, JJ, RB, DT, IN, .]

- Exhaustive search
    - For each token, we have to estimate the probability w.r.t. each tag.
    - $O(|T|^{|S|})$
    - Very expensive!!
- If we know that a word can take only a handful of PoS tags, we can reduce this search.
    - How to get this information?
        - Corpus
    - Still, it can have lots of possibilities.
        - Retain only the most probable path so far, and discard others.
        - Viterbi Algorithm

	0	1	2	3	4	5	6	7
W:	^	Brown	foxes	jumped	over	the	fence	.
T:								

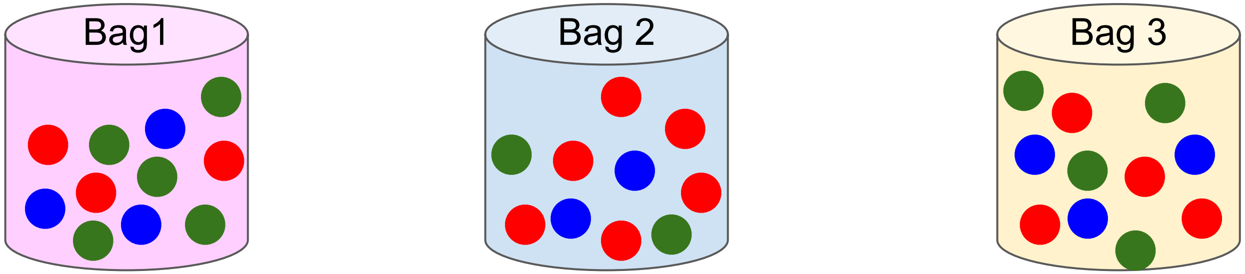




- **Pick a path to the leaf that maximizes the likelihood of the tag sequence.**
  - **E.g.,** *the top most path,*
    - **^ JJ NNS VBD NN DT NN .**

Hidden Markov Model (HMM)

Motivation



Bag1	Bag 2	Bag 3
R → 30	R → 10	R → 60
G → 50	G → 40	G → 10
B → 20	B → 50	B → 30

- Assume, we have an observation of N balls withdrawn from these bags

Red	Red	Green	Green	Blue	Red	Green	Red
B1, B2, or B3?	B1, B2, or B3?	B1, B2, or B3?			…		B1, B2, or B3?

- Question
  - Give the most likely sequence of bags for the withdrawal of above sequence of balls.
    - Not an easily computable problem.

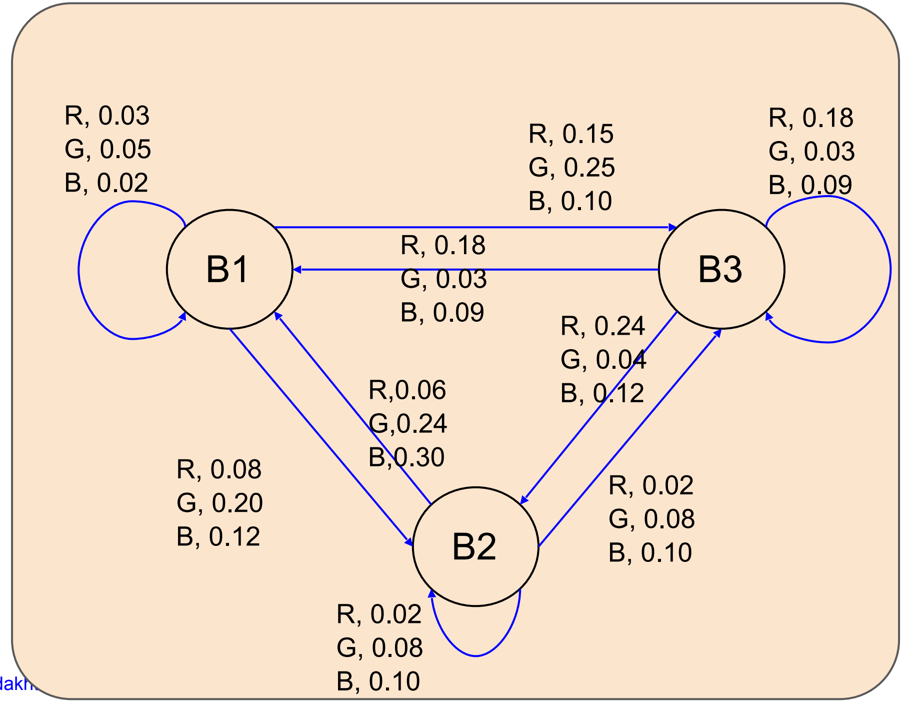
Given two probability matrices

Transition probabilities

	B1	B2	B3
B1	0.1	0.4	0.5
B2	0.6	0.2	0.2
B3	0.3	0.4	0.3

Emission probabilities

	R	G	B
B1	0.3	0.5	0.2
B2	0.1	0.4	0.5
B3	0.6	0.1	0.3



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Given two probability matrices

Transition probabilities

	B1	B2	B3
B1	0.1	0.4	0.5
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B3	0.3	0.4	0.3

Emission probabilities

	R	G	B
B1	0.3	0.5	0.2
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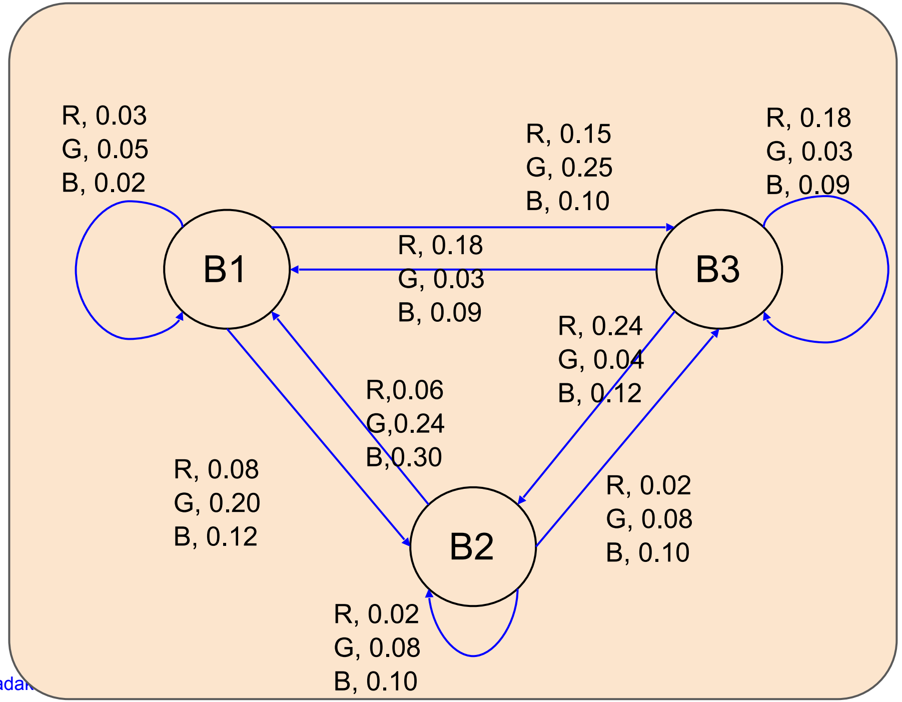


^	R	R	G	
Initial State B0	B1, 0.3*1 =0.3	B1, 0.3*0.1*0.3 =0.009	B1, 0.009*0.1*0.5	0.00045
			B2, 0.009*0.4*0.4	0.00144
			B3, 0.009*0.5*0.1	0.00045
		B2, 0.3*0.4*0.1 =0.012	B1, 0.012*0.6*0.5	0.0036
			B2, 0.012*0.2*0.4	0.00048
			B3, 0.012*0.2*0.1	0.00024
		B3, 0.3*0.5*0.6 = 0.09	B1, 0.09*0.3*0.5	0.0108
			B2, 0.09*0.4*0.4	0.0144
			B3, 0.09*0.3*0.1	0.0027

Formulation of HMM-based PoS tagging

	o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8
Observation	R	R	G	G	B	R	G	R
State	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8

where,

$s_i \in [B_1, B_2, B_3]$

$S*$: Best possible state sequence

Goal: Maximize $P(S \mid O)$ tag sequence by choosing the best sequence S

$$S* = argmax_S \, P(S \mid O)$$

Formulation of HMM

$$S^* = argmax_S \, P(S \mid O)$$

$$P(S \mid O) = P(\{s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8\} \mid \{o_1, o_2, o_3, o_4, o_5, o_6, o_7, o_8\})$$

1. Apply chain-rule

$$= P(s_1 \mid O)P(s_2 \mid s_1, O)P(s_3 \mid s_1 s_2, O)P(s_4 \mid s_1 s_2 s_3, O) \cdots P(s_8 \mid s_1 s_2 \cdots s_7, O)$$

2. Apply Markov assumption

$$= P(s_1 \mid O)P(s_2 \mid s_1, O)P(s_3 \mid s_2, O)P(s_4 \mid s_3, O) \cdots P(s_8 \mid s_7, O)$$

Bayes' Theorem

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

where,

$P(A \mid B)$	= Posterior
$P(B \mid A)$	= Likelihood
$P(A)$	= Prior
$P(B)$	= Normalizing constant

Formulation of HMM

3. Apply Bayes' theorem

$$S^* = argmax_S \; P(S \mid O) = argmax_S \; P(O|S) \,.\, P(S)$$

Prior

$$P(S) = P(s_1) \,.\, P(s_2|s_1) \,.\, P(s_3|s_2) \ldots P(s_8|s_7)$$

Likelihood

$$P(O \mid S) = P(o_1 \mid S)\, P(o_2 \mid o_1, S)\, P(o_3 \mid o_2, S)\, P(o_4 \mid o_3, S) \ldots P(o_8 \mid o_7, S)$$

4. Ball withdrawal depends on the current bag/state only.

$$P(O \mid S) = P(o_1 \mid s_1)\, P(o_2 \mid s_2)\, P(o_3 \mid s_3)\, P(o_4 \mid s_4) \ldots P(o_8 \mid s_8)$$

Putting prior and likelihood together

$$P(S \mid O) = P(O|S) \,.\, P(S)$$

$$= P(o_1 \mid s_1)\, P(o_2 \mid s_2)\, P(o_3 \mid s_3)\, P(o_4 \mid s_4) \ldots P(o_8 \mid s_8) \;\; P(s_1) \,.\, P(s_2|s_1) \,.\, P(s_3|s_2) \ldots P(s_8|s_7)$$

Formulation of HMM

	o_0	o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8	
Observation	ε	R	R	G	G	B	R	G	R	
State	s_0	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8	s_9

We introduce two new states, s_0 and s_9, to represent the initial and final states with ε symbol represents the start of the observation.

$$
\begin{aligned}
P(S \mid O) \quad = \quad & [P(o_0 \mid s_0)\, P(s_1|s_0)] \,. \\
& [P(o_1 \mid s_1)\, P(s_2|s_1)] \,. \\
& [P(o_2 \mid s_2)\, P(s_3|s_2)] \\
& [P(o_3 \mid s_3)\, P(s_4|s_3)] \\
& [P(o_4 \mid s_4)\, P(s_5|s_4)] \\
& [P(o_5 \mid s_5)\, P(s_6|s_5)] \\
& [P(o_6 \mid s_6)\, P(s_7|s_6)] \\
& [P(o_7 \mid s_7)\, P(s_8|s_7)] \\
& [P(o_8 \mid s_8)\, P(s_9|s_8)]
\end{aligned}
$$

$$P(S \mid O) \quad = \quad \prod_{k=1} P(o_k \mid s_k)\, P(s_{k+1}|s_k)$$

$$P(S \mid O) \quad = \quad \prod_{k=1} P(s_k \xrightarrow{o_k} s_{k+1})$$

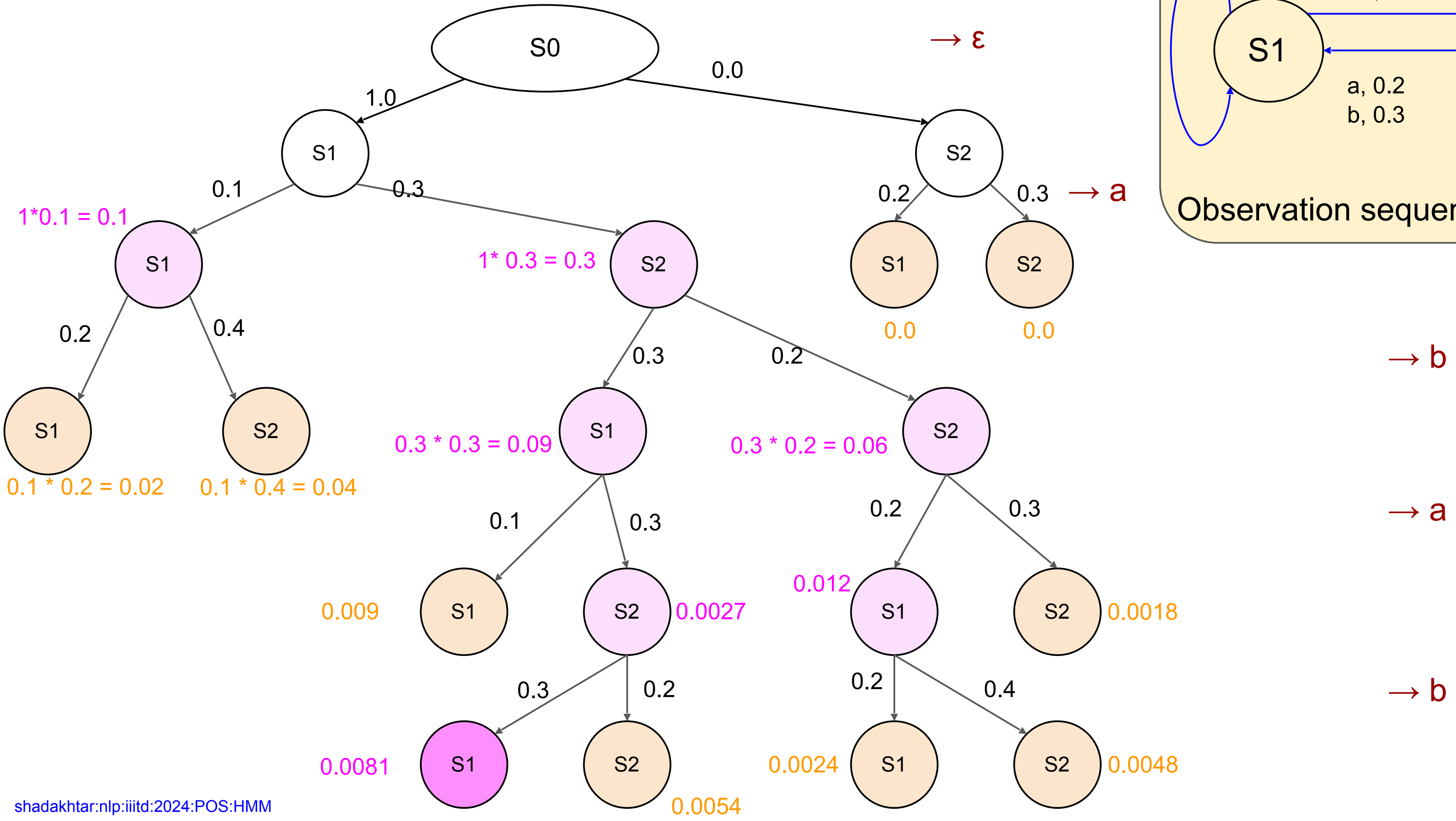
where,

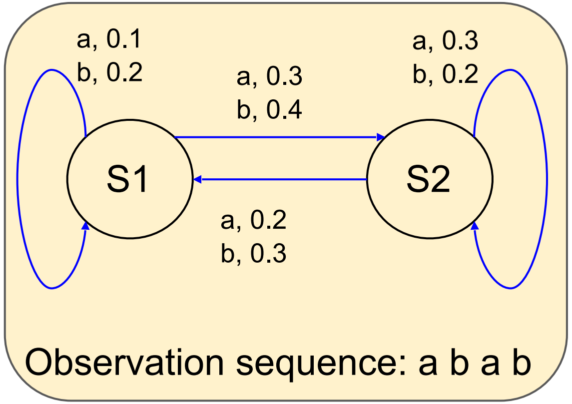
$P(s_9|s_8) = 1$, is the transition probability from the state of the last observation s_8 to the final state s_9

$P(s_1|s_0)$ is the initial transition probability.

$P(o_0 \mid s_0)$ is the initial emission probability

Decoding a state sequence





Summary of HMM

- Problem definition:
    - $S^* = argmax_S \; P(S \mid O, M)$ where,
        - $S \rightarrow$ State sequence,
        - $O \rightarrow$ Observation sequence,
        - $M \rightarrow$ Model
            - M = [Q, Q_0, A, T] where,
                - $Q \rightarrow$ Set of states,
                - $Q_0 \rightarrow$ Start state,
                - $A \rightarrow$ Alphabet, and
                - $T \rightarrow$ Transition function

$$P(s_i \xrightarrow{a_k} s_j) \quad \forall_{i,\,k,\,j}$$

Computing P(.) values for PoS tagging

- States (s_k) are the tags (NN, JJ, VB) and
- Observations (o_k) are the words in a sentence.

$$P(S \mid O) \quad = \quad \prod_{k=1} P(o_k \mid s_k)\, P(s_{k+1} \mid s_k)$$

$$P(NN \mid JJ) = \frac{Number\_of\_times\_JJ\_followed\_by\_NN}{Number\_of\_times\_JJ\_appeared}$$

$$P(Brown \mid JJ) = \frac{Number\_of\_times\_Brown\_tagged\_as\_JJ}{Number\_of\_times\_JJ\_appeared}$$

Associated tasks with HMM

- Given an automaton and an observation sequence, find the best possible state sequences
  - $S*$ = ?
  - Viterbi
- Given an automaton, find the probability of an observation sequence
  - $P(O)$ = ?
  - Forward algorithm: $F(k,i) = P(o_1 o_2 o_3 \cdots o_k, S_i) = \sum_{j=0,N} F(k-1,j) \,.\, P(s_j \rightarrow^{o_k} s_i)$
    - Probability of being in state s_i having seen $o_1 o_2 o_3 \cdots o_k$
  - Backward algorithm: $B(k,i) = P(o_k o_{k+1} o_{k+2} \cdots o_m, s_i) = \sum_{j=0,N} B(k+1,j) P(s_i \rightarrow^{o_k} s_j)$
    - Probability of seeing $o_k o_{k+1} o_{k+2} \cdots o_m$ given that the state was s_i
- Given the observation sequence, find the HMM parameters
  - $M(T)$ = ?
  - Baum-Welch algorithm or Forward-Backward algorithm: EM

Thanks